Skills Summary

Composite Rectilinear Figures: Area and Perimeter

Use this reference while completing your worksheet or when studying.

A **composite rectilinear figure** is a shape made only of rectangles, with every corner a square corner. You never need a new formula for one — you only need to decide *where to cut it*, and whether the question is asking about the *inside* (area) or the *edge* (perimeter).

1. Cut it into rectangles and add (composite-area-decompose)

Rule: Area of a rectangle = length $\times$ width, counted in *square* units.

Draw one straight line across the shape so it becomes two rectangles. Find each area, then **add**. The cutting line is never part of the answer — it is only there to help you see the rectangles.

Example: An L-shape: the bottom part is 6 across and 2 tall; sitting on its left end is a part 3 across and 3 tall (so the whole left side is 5).
Step 1. The shape is not a rectangle, so cut straight across at the top of the bottom part — now there are two rectangles and I can use length $\times$ width on each.
Step 2. Bottom rectangle: $6 \times 2 = 12$. Top rectangle: $3 \times 3 = 9$. $\Rightarrow$ $12 + 9 =$ **21** square centimetres

Try it: An L-shape has a bottom part 7 across and 2 tall, with a part 4 across and 4 tall sitting on its left end. Find the area in square centimetres.

check: **30**

2. Fill in the corner, then take it away (composite-area-subtract)

Rule: If the shape looks like a rectangle with one corner bitten out:

area = (whole rectangle) − (missing corner)

Use this when the missing corner is easy to see. It gives the same answer as cutting and adding, so pick whichever is less work — and use the other one to check.

Example: A rectangle 8 across and 5 tall has a corner piece 2 across and 3 tall removed.
Step 1. The bite is a whole rectangle by itself, so it is quicker to find the big area and subtract it than to cut the leftover shape up.
Step 2. Whole: $8 \times 5 = 40$. Missing corner: $2 \times 3 = 6$. $\Rightarrow$ $40 - 6 =$ **34** square centimetres

Try it: A rectangle 9 across and 6 tall has a corner piece 3 across and 2 tall removed. Find the area in square centimetres.

check: **48**

3. Walk all the way around (composite-perimeter)

Rule: Perimeter is the distance *around the edge*, so **add every side**, in plain units — never multiply.

Start at one corner, put your finger on each side in order, and tick it off as you add it. An L-shape has **six** sides, not four: the two short sides inside the notch count too.

Example: Going around an L-shape the sides measure 7, 3, 4, 2, 3 and 5.
Step 1. The question asks how far it is around the edge, so this is perimeter — add the sides, and count how many sides there should be first.
Step 2. Six sides, so six numbers: $7 + 3 + 4 + 2 + 3 + 5$. $\Rightarrow$ **24** centimetres

Try it: Going around an L-shape the sides measure 8, 2, 5, 4, 3 and 6. Find the perimeter in centimetres.

check: **28**

4. Sides that line up must match (missing-side)

Rule: In a rectilinear figure, the short pieces along one direction add up to the long side facing them:

short + short = long, so missing = long − known short.

Horizontal pieces match the horizontal side; vertical pieces match the vertical side.

Example: The bottom of a shape is 11 long. Along the top, one piece is 4 and the other is unknown.
Step 1. The two top pieces together stretch exactly as far as the bottom, so this is a subtraction, not a guess.
Step 2. $4+? = 11$, so $? = 11 - 4$. $\Rightarrow$ **7** centimetres

Try it: The left side of a shape is 10 tall. On the right, the upper piece is 6 tall and the lower piece is unknown. How tall is the lower piece, in centimetres?

check: **4**

(!) Watch out: *Area* is covered space and is counted in **square** units (cm^2, ft^2); *perimeter* is distance around and is counted in plain units (cm, ft). Underline the word “area” or “perimeter” in the question before you pick up your pencil, and never leave out the two hidden sides inside a notch.